

Parametric Retrieval Augmented Generation

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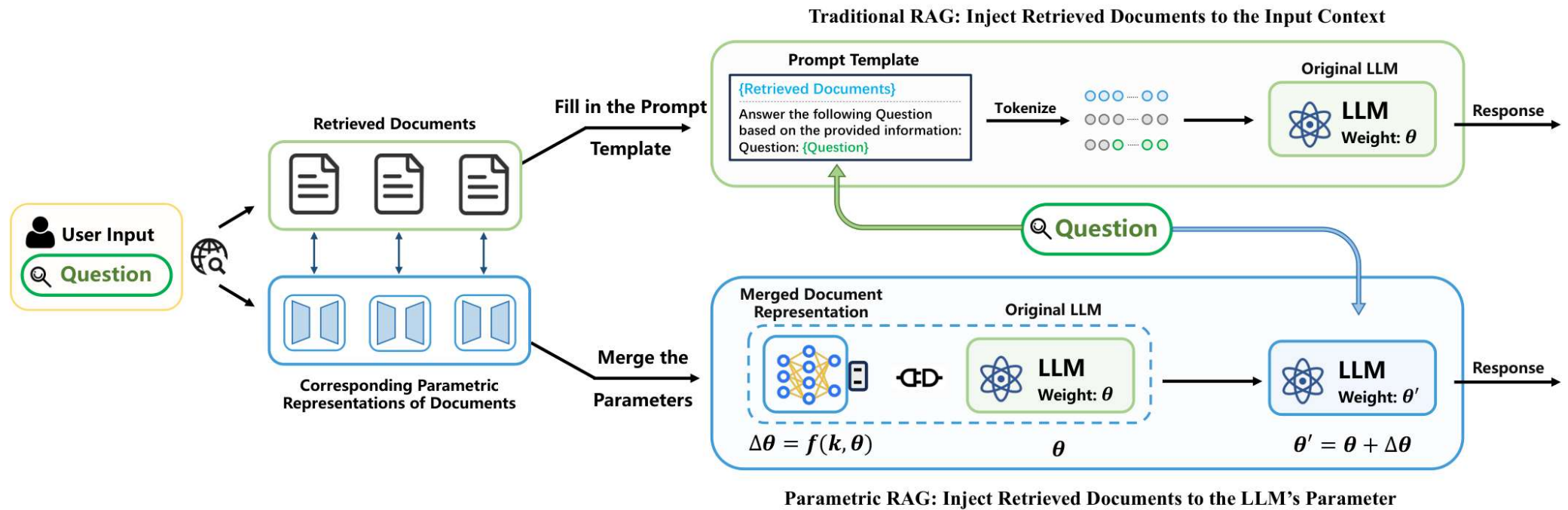
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- inject external knowledge by adding into the input context
 - introduce extra computational overhead and **latency**
 - LLMs store most of **knowledge** within the **parameters**
 - only affect the online computation of **key-value (KV) pairs**, but not the model's stored parameters

Methodology

- Offline Document Parameterization
- Online Inference



Offline Document Parameterization

- Document Augmentation
 - Document Rewriting
 - prompt the LLM to rewrite the content using different wording, styles, or organizational structures
 - QA Pair Generation
 - prompt LLM to generate question-answer (QA) pairs

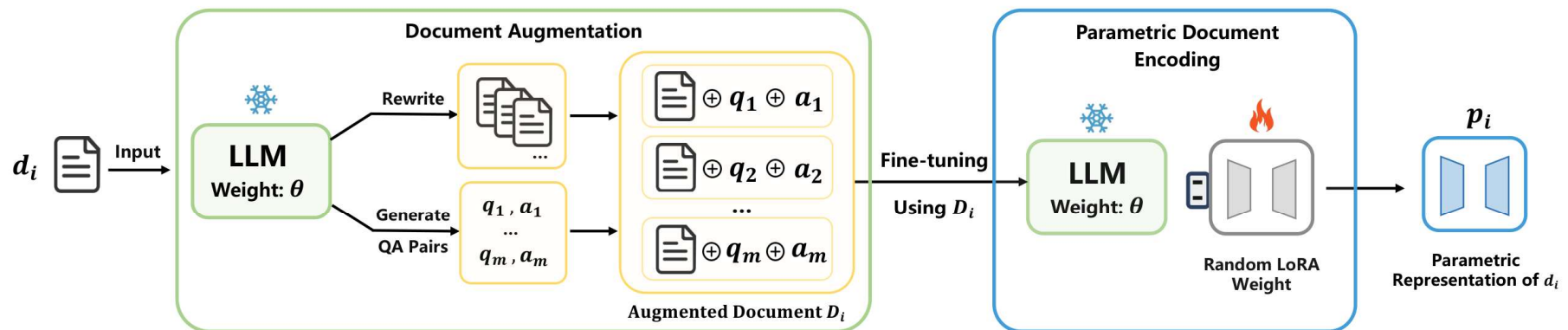


Figure 2: An illustration of how we parameterize each document d_i in the corpus during the *Offline Document Parameterization* stage.

Offline Document Parameterization

- Parametric Document Encoding
 - FFN layer
 - LoRA approach
 - few-shot examples and save the LoRA weight W_{warm-}

$$W' = W + \Delta W = W + AB^T, \quad \min_{\Delta\theta} \sum_{(d_i^k, q_i^j, a_i^j) \in D_i} \sum_{t=1}^T -\log P_{\theta+\Delta\theta}(x_t | x_{<t}),$$

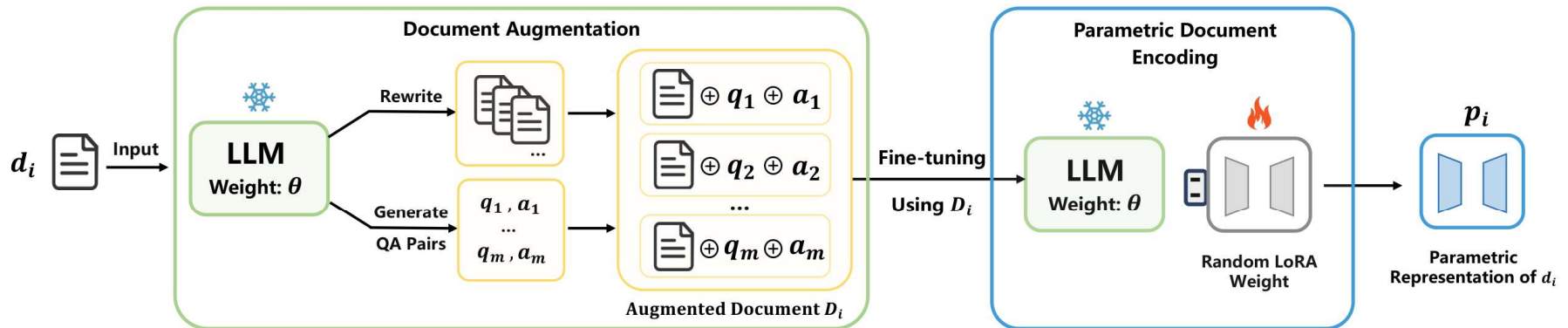


Figure 2: An illustration of how we parameterize each document d_i in the corpus during the *Offline Document Parameterization* stage.

Online Inference

- Retrieve

- Select top-k document

- Update

- Use a scalar scaling factor α to modulate the final update

$$\Delta W_{\text{merge}} = \alpha \cdot \sum_{j=1}^k A_j B_j^{\top}.$$

- Generate

- obtain a temporary model $L'(\theta')$

Experimental Setup

- **Models:**

- Qwen2.5-1.5B-Instruct , LLaMA-3.2-1B-Instruct, Llama-3-8B-Instruct

- **Baselines:**

- Standard RAG, DA-RAG, FLARE, DRAGIN, P-RAG, Combine Both

- **Dataset:**

- 2WikiMultihopQA (2WQA), HotpotQA (HQA), PopQA (PQA), ComplexWebQuestions (CWQ)

- **Metrics:** F1 score

- **Hardware:** NVIDIA A100 GPUs with 40GB

Results

Table 1: The overall experiment results of Parametric RAG and other baselines across four tasks. All metrics reported are F1 scores. Bold numbers indicate the best performance of all baselines, and the second-best results are underlined. “*” and † denote significantly worse performance than the bolded method and our proposed P-RAG with $p < 0.05$ level, respectively.

		2WikiMultihopQA					HotpotQA			PopQA	CWQ
		Compare	Bridge	Inf.	Compose	Total	Bridge	Compare	Total		
LLaMA-1B	Standard RAG	0.4298 ^{†*}	0.3032 ^{†*}	0.2263	0.1064	0.2520 [*]	<u>0.2110</u>	0.4083	<u>0.2671</u>	0.1839 [*]	0.3726
	DA-RAG	0.3594 ^{†*}	0.2587 ^{†*}	<u>0.2266</u>	0.0869 ^{†*}	0.2531 [*]	0.1716 [*]	0.3713 ^{†*}	0.2221	0.2012 [*]	0.3691
	FLARE	0.4013 ^{†*}	0.2589 ^{†*}	0.1960	0.0823 ^{†*}	0.2234 [*]	0.1630 [*]	0.3784 ^{†*}	0.1785 [*]	0.1301 ^{†*}	0.3173 [*]
	DRAGIN	0.4556	0.3357 [*]	0.1919	0.0901	0.2692 [*]	0.1431 [*]	0.4015	0.1830 [*]	0.1056 ^{†*}	<u>0.3900</u>
	P-RAG (Ours)	<u>0.4920</u>	<u>0.3994</u>	0.2185	<u>0.1334</u>	<u>0.2764</u>	0.1602 [*]	0.4493	0.1999 [*]	<u>0.2205[*]</u>	0.3482 [*]
	Combine Both	0.5046	0.4595	0.2399	0.1357	0.3237	0.2282	<u>0.4217</u>	0.2689	0.2961	0.4101
Qwen-1.5B	Standard RAG	0.3875 ^{†*}	0.3884 ^{†*}	0.1187 ^{†*}	0.0568 ^{†*}	0.2431 ^{†*}	0.1619 [*]	0.3713 ^{†*}	0.2073 [*]	0.0999 ^{†*}	0.2823 [*]
	DA-RAG	0.3418 ^{†*}	0.4015	0.1269 ^{†*}	0.0514 ^{†*}	0.2156 ^{†*}	0.1182 ^{†*}	0.3041 ^{†*}	0.1683 [*]	0.1197 ^{†*}	0.2718 ^{†*}
	FLARE	0.1896 ^{†*}	0.1282 ^{†*}	0.0852 ^{†*}	0.0437 ^{†*}	0.1004 ^{†*}	0.0750 ^{†*}	0.1229 ^{†*}	0.0698 ^{†*}	0.0641 ^{†*}	0.1647 ^{†*}
	DRAGIN	0.2771 ^{†*}	0.1826 ^{†*}	0.1025 ^{†*}	0.0680 ^{†*}	0.1538 ^{†*}	0.0801 ^{†*}	0.1851 ^{†*}	0.0973 ^{†*}	0.0548 ^{†*}	0.1788 ^{†*}
	P-RAG (Ours)	0.4529	0.4494	0.2072	0.1372	0.3025	<u>0.1720</u>	<u>0.4623</u>	<u>0.2165[*]</u>	<u>0.1885</u>	<u>0.3280</u>
	Combine Both	<u>0.4053</u>	<u>0.4420</u>	<u>0.1705</u>	<u>0.1154</u>	<u>0.2627</u>	0.2383	0.5037	0.2942	0.2261	0.3495
LLaMA-8B	Standard RAG	0.5843 ^{†*}	0.4794 ^{†*}	0.1833 ^{†*}	0.0991 ^{†*}	0.3372 ^{†*}	0.1823 ^{†*}	0.3493 ^{†*}	0.2277 ^{†*}	0.1613 ^{†*}	0.3545 ^{†*}
	DA-RAG	0.4921 ^{†*}	0.3344 ^{†*}	0.1523 ^{†*}	0.0670 ^{†*}	0.2396 ^{†*}	0.1587 ^{†*}	0.2860 ^{†*}	0.1996 ^{†*}	0.2255 [*]	0.3481 ^{†*}
	FLARE	0.4293 ^{†*}	0.3769 ^{†*}	<u>0.3086</u>	0.1627 [*]	0.3492 [*]	0.2493 ^{†*}	0.4324 ^{†*}	0.2771 ^{†*}	0.2393 [*]	0.3084 ^{†*}
	DRAGIN	0.5185 ^{†*}	0.4480 ^{†*}	0.2664	0.1833	0.3544 [*]	0.2618 [*]	0.6116 [*]	0.2924 [*]	0.1772 ^{†*}	0.3101 ^{†*}
	P-RAG (Ours)	<u>0.6353</u>	<u>0.5437</u>	0.2471 [*]	<u>0.1992</u>	<u>0.3932</u>	<u>0.3115[*]</u>	<u>0.6557</u>	<u>0.3563[*]</u>	<u>0.2413[*]</u>	<u>0.4541</u>
	Combine Both	0.6432	0.5556	0.3160	0.2339	0.4258	0.4025	0.6918	0.4559	0.3059	0.4728

Results

Table 2: Ablation study on the impact of LoRA weight initialization strategies for P-RAG. All metrics reported are F1 scores. “P-RAG Rand.” and “P-RAG Warm.” indicate randomly initialized LoRA weights and warm-up LoRA initialization, respectively. The best results are in bold.

		2WQA	HQA	PQA	CWQ
LLaMA-1B	P-RAG Rand.	0.2764	0.1999	0.2205	0.3482
	P-RAG Warm.	0.3546	0.2456	0.2035	0.4263
Qwen-1.5B	P-RAG Rand.	0.3025	0.2165	0.1885	0.3280
	P-RAG Warm.	0.3542	0.2718	0.2418	0.5018
LLaMA-8B	P-RAG Rand.	0.3932	0.3563	0.2413	0.4541
	P-RAG Warm.	0.4201	0.4499	0.2952	0.5591

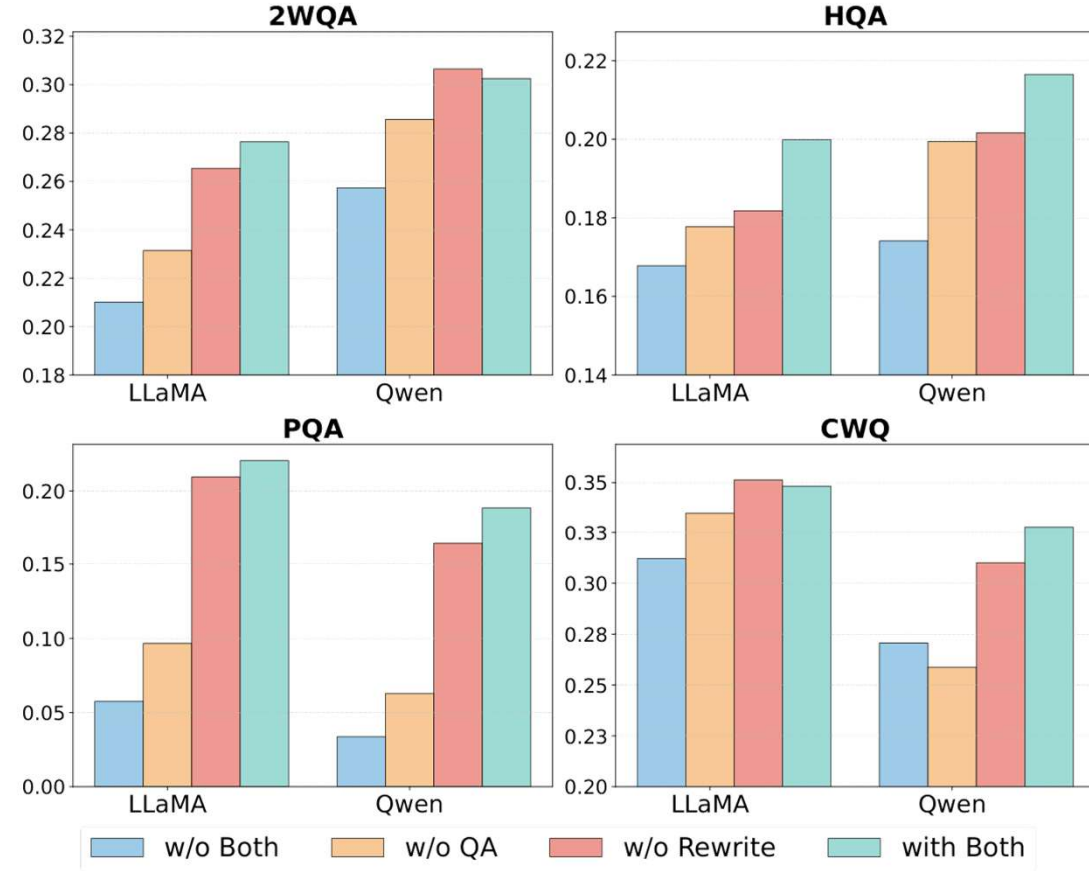


Figure 3: Ablation study on the impact of the document augmentation stage. LLaMA indicates LLaMA-3.2-1B, and Qwen indicates Qwen-2.5-1.5B. The metric used is the F1 Score@.

Results

Table 3: Ablation study comparing different document augmentation models. GenLM indicates the generator LLM and AugLM indicates the LLM for document augmentation. LLaMA indicates LLaMA-3.2-1B, and Qwen indicates Qwen-2.5-1.5B. The best results are in bold. The metric used in the table is F1 Score.

GenLM	AugLM	Dataset			
		2WQA	HQA	PQA	CWQ
LLaMA-1B	LLaMA-1B	0.2764	0.1999	0.2205	0.3482
	Qwen-1.5B	0.2753	0.1980	0.2340	0.3495
	LLaMA-8B	0.2748	0.1935	0.2207	0.3498
Qwen-1.5B	LLaMA-1B	0.2974	0.2005	0.1829	0.3183
	Qwen-1.5B	0.3025	0.2165	0.1885	0.3280
	LLaMA-8B	0.2948	0.2161	0.2156	0.3211

Table 4: The average time required by the LLaMA3-8B model to answer a question on the 2WikiMultihopQA (2WQA) and ComplexWebQuestions (CWQ) datasets. The "+0.32" footnote for P-RAG and Combine Both indicates the total time needed for merging and loading the LoRA adapter.

	2WQA		CWQ	
	Time(s)	Speed Up	Time(s)	Speed Up
P-RAG	2.34 _{+0.32}	1.29x	2.07 _{+0.32}	1.36x
Combine Both	3.08 _{+0.32}	0.98x	2.84 _{+0.32}	0.99x
Standard RAG	3.03	1.00x	2.82	1.00x
FLARE	10.14	0.25x	11.31	0.25x
DRAGIN	14.60	0.21x	16.21	0.17x

Summary

Problem

- Adding external knowledge through the input context causes latency
- Fail to incorporate into parameters

Solution

- Offline Document Parameterization
- Online Inference

Result

- Parametric RAG offers a more efficient and scalable pathway to integrate external knowledge into LLMs